Geode: A Spontaneous Location-Based Online Social Network

Introduction

In today's digital world, social media dominates the way that people explore, experience, and connect with others. However, most platforms encourage users to passively consume content rather than interacting with their environment. This project, *Geode*, uses augmented reality elements to merge exploration and digital interaction. Geode ties real word locations to photo challenge pins with prompts like "take a photo upside down" or "capture a moment with a new friend." These challenges can only be unlocked and completed by visiting the specified location, making exploration, creativity, and competition an integral part of the platform.

The motivation for this project stems from my personal experience while at DKU. Many of my peers travel frequently both within China, and in Asia more broadly, seeking memorable and unique experiences. We noticed that while platforms like Instagram allow us to share highlights, they lack the gamification and location-based mechanics that could make exploration more interesting in real-time. By combining elements of geolocation, social gaming, and photo sharing, Geode seeks to create a system where people discover hidden places, foster social interaction, and compete in fun and meaningful ways.

This project will also contribute to my Computer Science degree by combining front-end and back-end development experience with UX design, and networked systems. More broadly, for the Computation and Design major, it will offer an opportunity to integrate technical experience with human-centered design, while reflecting on the broader social implications of gamifying travel and discovery.

Components

The project will consist of several interrelated components:

1. Core Application Design

Geode's foundation lies in its mobile application, built using a MERN (MongoDB, Express, React, Node.js) stack with Firebase authentication. The initial prototype leverages React Native for cross-platform compatibility, ensuring usability for both iOS and Android users. The architecture allows users to drop "challenges" tied to geographic coordinates, which then become accessible to others visiting that location.

2. Gamification and Social Mechanics

The project incorporates game design mechanics to create engagement loops. Challenges are competitive, and participants' submissions can be ranked or voted on by peers. This draws inspiration from viral phenomena such as *Pokémon Go*, which demonstrated measurable impacts on physical activity and social interaction (Khamzina, Parab, Kaustubh V, An, Bullard, &

Grigsby-Toussaint, 2020). Unlike Pokémon Go's focus on collection, Geode incorporates user-generated content, keeping the game fresh.

This gamification element draws on insights from Bulchand-Gidumal (2024), who studied how spontaneous online social networks (SOSNs) like BeReal impacted tourism. Similarly, Geode encourages users to capture spontaneous, place-based experiences rather than polished, curated content.

3. User Experience and Cultural Discovery

Geode is not just a game but also a tool for uncovering unique cultural and geographic experiences. By allowing travelers or locals to plant challenges in overlooked places, the app fosters grassroots discovery of "special spots" that are not captured by traditional tourism platforms like Google Maps. This dynamic aligns with broader trends in tourism innovation, where gamified and participatory platforms drive new forms of engagement, especially among younger generations (Skinner, Sarpong, & White, 2018).

4. Potential Future Improvements and Scalability

While this project's technical foundation relies on the scalable, well-documented MERN stack, future versions of Geode could explore AI-assisted moderation (e.g., detecting inappropriate content in photos) and recommendation systems for suggesting challenges to users. The feasibility of these possibilities will also be evaluated.

One of the unique aspects of Geode is its reliance on network effects. The platform becomes more valuable as more users create and participate in challenges. This design decision reflects research into social computing and viral adoption patterns (Reichstein & Brusch, 2019).

Conclusion

The *Geode* project represents an interdisciplinary exploration of how computation, design, and social play can intersect to create meaningful experiences. Technically, it challenges me to build a robust and scalable system, integrating full-stack development, geolocation services, and user experience design. Creatively, it invites me to consider how gamification and cultural discovery can be thoughtfully incorporated into digital platforms. Academically, it allows me to connect key themes from the Computation and Design major—human-computer interaction, system design, and computational media—with real-world social applications.

Ultimately, I expect Geode to demonstrate how computation and design can inspire more active, social, and creative engagement with our environments. Whether it becomes a platform for travelers, students, or local communities, the project reflects the broader question at the heart of my academic path: how can computational systems be designed not just to process data, but to shape the ways we explore and experience the world?

Sources

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Your individual project title (Notes: Please format your individual project title as the following "Discipline: Contextual SW Project title.")*

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Your team project title if applicable (Notes: Please format your team project title as the following "Discipline: Contextual SW Project title.")

Computation and Design: Gamification and Place-Based Social Media

Key Words

1. **Gamification**
2. **Geolocation**
3. **Social Networking**
4. **Exploration**
5. **User Engagement**

Why do you want to change your SW topic? (200 words)*

There are several reasons. Primarily, we found it difficult for people to be willing to speak on the record about semiconductors, especially in China and Taiwan, a segment critical to our project. With this missing, the report and interactive display would not be able to be complete, or at the very least, areas critical to the manufacture and trade of semiconductors would be missing from our analysis. Furthermore, Colden and I have had experiences at our summer internships that, rather than veering off in a semiconductor industry analysis direction, better suit us to work on the aspects of this project. Finally, producing *Geode* more closely aligns with our Computation and Design major and Computer Science track, allowing us to consider the implications of place-based social media, gamification, user engagement, and social networks more broadly in a way that a semiconductor industry survey would not. It will provide more exposure to app development which is more related to a Computer Science track.